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Solar irradiance resource and forecasting: a comprehensive review

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Abstract: With the increase in demand for energy, penetration of alternative sources of energy in the power grid has increased. Photovoltaic (PV) energy is the most common and popular form of energy sources which is widely integrated into the existing grid. As solar energy is intermittent in nature, to ensure uninterrupted and reliable power supply to the prosumers, it is essential to forecast the solar irradiance. Accurate solar forecasting is necessary to facilitate large-scale modelling and deployment of PV plants without disrupting the quality and reliability of the power grid as well as to manage the power demand and supply. There are various methods to predict the solar irradiance such as numerical weather prediction methods, satellite-based approaches, cloud-image based methodologies, data-driven methods, and sensor-network based approaches. This study gives an overall review of the different resources and methods used for forecasting solar irradiance in different time horizons and also gives an extensive review of the sensor networks that are used for determining solar irradiance. The various error metrics and accessible data sets available for the sensor networks are also discussed that can be used for validation purposes.

1 Introduction

Factors like global warming and energy crisis have increased the usage of alternative sources of energy in the last few decades, owing to their properties of being sustainable and clean. Among these sources of energy, solar energy has gained the utmost popularity as it is inexhaustible and considered to be the most promising renewable energy resource for power generation on a large scale. One of the underlying technologies used for converting solar energy into electricity comprises photovoltaic (PV) cells [1]. PV energy is getting penetrated into the power grid extensively in the recent years, both on a large scale as well as small scale [2]. Being intermittent in nature, PV power introduces significant volatility to the grid, which in turn affects the stability of the grid [3]. This potentially unpredictable grid instabilities affect the adoption and acceptance of PV power by the utility companies and the industries [4]. When the PV power is unpredictable, the utilities tend to purchase additional power or over-produce electricity. The effects of such intermittencies and fluctuations exhibited by PV can be reduced by employing energy storage, demand response, and spinning reserves. However, for optimal usage of these methods, forecasting of solar irradiance is mandatory.

Various methods are being used by researchers and professionals for forecasting the solar irradiance, which can be broadly categorised as data-driven approaches, image-based approaches, numerical weather prediction (NWP) models, and hybrid approaches. Data-driven approaches make use of the historical data that have been measured or sensed earlier, to forecast the solar time series. Image-based approaches use images from sky/shadow cameras or satellites, combine it with historical data and eventually forecast the irradiance. NWP models use

computer-based programs to predict the irradiance. Hybrid or combinational models are different combinations of two or more of the above-mentioned models.

The forecast horizon is vital from the application point of view. Depending on the need of forecast horizon stretch, methods/models are adopted by the system. Based on the decision-making activities in smart grid, the forecast horizon can be classified as very short-term, short-term, medium-term, and long-term [1], which may vary depending on the usage and assumption by different researchers. The applications of solar forecasting are quite diverse when applied to smart grid. Forecasts from a few seconds to a few minutes are used to control rapid fluctuations and ramp rates, which in turn results in smooth power production, real-time dispatch, accurate medium and long-term dispatch, appropriate reserve allocation etc. In general, daily and hourly fluctuations are taken care by solar forecasting for a few hours in advance, and such forecasts are used for unit commitment, economic dispatch etc. Moreover, weekly forecasts are used for maintenance scheduling, and monthly/annual forecasts are used for planning of PV power plants.

Selecting a method for forecasting the PV power or irradiance mainly depends on the application area and the forecast horizon. A method can have different efficiency in the same horizon, depending on the data set used. It may happen that a method over-performs the other in a particular forecast horizon, but is less efficient in some other forecast horizon. The authors of [5] discussed that cloud motion vectors (CMVs) provide better results than NWP for forecasts up to 3–4 h. Beyond this horizon of a few hours, NWP models perform better and give more accurate results. In [6], the researchers have used mean bias errors (MBEs) and proved with experimental results that cloud motion forecasts stand out persistence forecasts which have been derived by performing actual measurements, up to 6 h after sunrise. In [7] a detailed discussion is provided on the selection of suitable methods for solar irradiance forecasting, depending on forecast horizons. Based on studies and researches done, a table is devised for suitability of methods for different forecast horizons, which is supported by [8–10]. Table 1 summarises this work.

This study gives a detailed review of the irradiance resource and methods available for solar irradiance forecasting with a focus on sensor networks. Besides, the error metrics, some of the sensor networks' data sets used in literature are also discussed. The main

Table 1 Forecasting horizon and preferred forecasting method

Forecasting horizon	Preferred forecasting method
1 min to 10 mins	persistence of ground measurements
10 mins to 1 h	ground based CMVs data-driven methods
1 h to 5 h	satellite-based CMVs
5 h to 10 days	NWP models

aim of this study is to make the readers aware of recent studies on forecasting the solar irradiance worldwide and also to get acquainted with the accessibility of data sets. The rest of the study is organised as follows: Section 2 discussed the fundamentals and terminologies involved in solar irradiance. Section 3 presents an overview of the different forecasting techniques in literature based on time series and meteorological data and also describes the various techniques that are applied to sensor networks for determining solar irradiance. Section 4 discusses the various data sets available for sensor networks and their accessibility. Section 5 describes the metrics assessment of deterministic irradiance forecasting and Section 6 concludes the work.

2 Solar irradiance fundamentals

In this section, the fundamental components, instruments and terminologies that are generally used in solar irradiance forecasting methods are described.

2.1 Components of solar irradiance

The total extraterrestrial beam irradiance (EBI) from sun incident on the atmosphere of the Earth is represented by I_0^0 , which decreases as it descends to surface of the earth because of absorption, reflections and re-emissions caused in the atmosphere. The EBI incident on the surface of the earth has two components namely, direct horizontal irradiance (DHI), and direct normal irradiance (DNI). The geometric sum of these two components can be termed as global horizontal irradiance (GHI), which is also described as the solar irradiance incident on a horizontal plane on the surface of the Earth [11] and is represented in (1) according to [12].

$$\text{GHI} = \text{DHI} + \text{DNI} \cos(\theta), \quad (1)$$

where θ is the solar zenith angle.

GHI can be described as the total aggregate of solar radiation incident on a horizontal surface. It is not only used in fixed PV installations, but also to perform measure-correlate-predict evaluations using comparisons with the solar database. GHI is generally measured using horizontal pyranometers and silicon reference cells.

Global tilted irradiation (GTI) accounts for the entire amount of solar radiation that is collected through a tilted surface, falling directly over it. It can be defined as an estimate value for calculating the energy yield of slanting PV panels. GTI is used in fixed PV installations and is measured using pyranometer aligned in a similar direction as the solar module and reference cells.

DNI can be described as the aggregate of solar radiation falling perpendicularly on a surface, following a linear path from the current position of the sun. It is used in concentrated PV (CPV), concentrated solar power (CSP) and fixed PV installations. Pyrheliometer and rotating shadowband irradiometer are used to measure the DNI.

DHI is the shadow-free solar radiation that is collected by a surface per unit area, which reaches the earth's surface omnidirectionally after getting scattered by particles in the atmosphere. It is used in fixed PV installations and redundancy calculations of GHI. It is measured using pyranometer, which is installed in a sun tracker and a rotating shadowband irradiometer.

2.2 Clear-sky index (CSI) and clearness index (CI)

CSI and CI are the two terms that are widely used in solar irradiance forecasting. The CSI (k_t) is described as the ratio of measured irradiance (I_t) to the calculated clear-sky irradiance (I_t^{clr}).

$$\text{CSI} = k_t = \frac{I_t}{I_t^{\text{clr}}}. \quad (2)$$

The CI (K_t) is described as the ratio of normalised irradiance at the ground level (I_t) to the extraterrestrial irradiance (I_t^{EX}).

$$\text{CI} = K_t = \frac{I_t}{I_t^{\text{EX}}}, \quad (3)$$

where $I_t^{\text{EX}} = I_0 \cdot \cos(\theta)_t$, and $I_0 = 1360 \text{ W/m}^2$.

2.3 Clear-sky models (CSMs)

CSMs form an important part of the solar irradiance forecasting. Absence of clouds characterise the clear sky conditions, which is required by many resource assessment applications. Radiative transfer models (RTMs) are used to design CSMs, and the inputs to these models include the regional meteorological factors such as, ozone content, water vapour content, and Linke turbidity coefficient [12]. In [13], Ineichen gave a very detailed comparison of eight CSMs, suggesting that model selection criteria should depend on the application and availability of the data and not on the model accuracy. Additionally, a recent study conducted a worldwide comparison of 75 CSMs against 75 ground stations covering major climate zones [14]. MAC2, REST2v5, and REST2v9.1 models are reported as the best performing CSMs globally. The frequently used CSMs include Kasten model, REST2 model, McClear model, CPC2 model, and Ineichen-Perez model.

2.4 Instruments

Due to the processes of scattering and absorption in the atmosphere, a small portion of the solar spectrum enters the Earth's surface, which makes it essential to measure these radiations efficiently. Few of the popularly used instruments employed for measurements are discussed below.

2.4.1 Pyranometers: A pyranometer is an instrument for measuring the solar energy coming into the system. A pyranometer can be installed as an individual identity or as a part of a meteorological station [15]. The pyranometers as a whole class offer certain advantages [16] like they have very small temperature coefficient, produce more accurate measurements of performance index, are calibrated to international organization for standardization (ISO) standards and performance ratio, have longer response time than a PVcell, and render integrated measurement of the total available short-wave solar energy irrespective of conditions. They are used to predict insulation requirements for building structures, measure solar intensity data, design PV systems, and for meteorological studies.

Different types of pyranometers are used for measuring solar irradiance. A thermopile pyranometer can measure the total amount of radiation on a surface. It has a thermopile detector (a device for converting thermal energy into electrical energy), having a coat of light-absorbing black paint for consuming the constituents of solar radiations. This arrangement creates a difference in the temperature between the body of an instrument and surface of the sensors (which are painted black), resulting in a small voltage at the sensor, which can be measured and converted into units of W/m^2 . Such pyranometers measure radiations in the range 300 to 2800 nm. The thermopile absorbs all the solar radiation i.e., 300 to 50,000 nm, but the glass dome limits it to the range 300 to 2800 nm. These pyranometers are installed horizontally, being mounted in the 'plane of array' in such a way that the surface of the sensor lies parallel to the solar panel, in the event of monitoring the PV systems. The next category is the photodiode or silicon or semiconductor pyranometer, which detects the solar radiation between 400 to 900 nm. A silicon pyranometer makes use of a photodiode (a device for converting light into current) resulting in an electrical signal from the incoming solar radiation. Such pyranometers are comprised of a photodiode, a housing dome, and a diffuser/optical filter. The photodiode behaves as a sensor and generates a current which is directly proportional to the irradiance. They are used to calculate certain portions of the visible solar spectrum such as ultraviolet, infrared or photo-synthetically active radiation. The disadvantage of silicon pyranometers is their limited spectral sensitivity. The third category of pyranometers called PV pyranometers is a successor of the photodiode pyranometer. It

adheres to the requirement for a single reference PV cell to measure the cell's power and PV modules. This type of pyranometer is also known as 'reference PV cell', 'irradiance sensor', 'solarimeter' or 'solar sensor'. These pyranometers are applied in simulators and with PV systems for calculating power and system performance.

As per World Meteorological Organization and the ISO, following ISO 9060 standards, the pyranometers are classified as first-class pyranometers, second-class pyranometers and secondary-standard pyranometers. The first-class pyranometers are considered to be of good quality and are used for measurements in hydrology networks and greenhouse climate control. Whereas the second-class pyranometers are of mediocre quality and used mainly for daily measurements in weather stations and in field testing. The secondary-standard pyranometers are considered to be of scientific quality possessing highest accuracy. They find their applications in meteorology and testing in PV/CPV/CSP. These three classes of the pyranometers can be distinguished on the basis of the following factors: (i) response time which is defined as time to reach 95% of the response rate, (ii) non-stability, which is defined as the percentage deviation in responsivity per year, (iii) non-linearity, which is defined as the percentage deviation from responsivity at 500 W/m² for a change in irradiance from 100 to 1000 W/m², (iv) directional response, which is defined as the range of errors occurring due to the assumption that the typical incidence responsivity has omni-dimensional validity when measured from any direction, (v) beam radiation having normal incidence irradiance as 1000 W/m², (vi) spectral selectivity, which is defined as the product change of spectral absorptency and transmittance from the corresponding mean, (vii) temperature response, (viii) tilt response (percentage of change in responsivity corresponding from 0 to 90° tilt at 1000 W/m² beam irradiance) and (ix) achievable uncertainty. The discrimination between the three classes concerning the above-defined parameters are shown in Table 2. Among these three classes, the secondary standard pyranometers are considered to be the best followed by first-class and second-class, respectively.

These pyranometers are preferred over silicon reference cells owing to various advantages like providing an independent, accurate reading of the total available solar radiation; longer response time; small temperature coefficient; less pollution; and more accurate performance ratio or performance index. These characteristics make pyranometers better than silicon reference cells.

2.4.2 Silicon reference cells: The standard crystalline silicon reference cell is a precision sensor of high quality, used to determine the irradiance levels in a solar simulator. It is a low-cost reference cell which can be used efficiently for indoor applications [17]. It has temperature response which is similar to that of the PV system (in order of milliseconds) and is linear in nature, which allows room for correction in the temperature response.

Table 2 Difference among three classes of pyranometers

ISO 9060 pyranometer specifications	Secondary-standard pyranometers	First-class pyranometers	Second-class pyranometers
time of response	< 15 s	< 30 s	< 60 s
non-stability	±0.8%	±1.5%	±3%
non-linearity	±0.5%	±1%	±3%
directional response	±10 W/M ²	±20 W/M ²	±20 W/M ²
spectral selectivity	±3%	±5%	±10%
temperature response	2%	4%	8%
tilt response	±0.5%	±2%	±5%
achievable uncertainty hourly	3%	8%	20%
daily	2%	5%	10%

Silicon reference cells are used to measure irradiance, making use of the photons. Photons that has energy over the bandgap of the PV material are converted directly into positive and negative charges. These charges are collected and used in an external circuit. These cells generate current, which is dependent on the number of photons and their spectral distribution. Silicon reference cells do not respond to photons with energy less than the bandgap.

In general, reference cells comprises a solar cell, a temperature sensor and a rugged enclosure with a protective window. Features of the most common configurations of reference cells include low thermal mass, temperature sensor (four-wire contact to the solar cell BK7 glass window), solid thermal contact between the solar cell and the enclosure, low-reflectivity coating trailing the window and enclosing solar cell, broad-spectrum and mounting hole pattern compatible with WPVS reference cell design. The operating temperature for this cell is −35°C to 80°C. Silicon reference cells can be categorised as crystalline and non-crystalline. The crystalline cells include mono-Si, multi-Si, and ribbon-Si. The non-crystalline cells include amorphous silicon (a-Si), non-crystalline silicon (nc-Si), protocrystalline silicon (pc-Si), emerging PVs and multi-junction reference cells.

2.4.3 Pyrheliometer: Pyrheliometer is an instrument used for measuring direct beam irradiance at normal incidence. Light from the sun enters the instrument through a window, which is then directed onto a thermopile. The thermopile converts solar heat into a recordable electrical signal. The signal voltage is used along a solar tracking device for keeping the pyrheliometer directed at the Sun. A pyrheliometer is often used in the same setup with a pyranometer [15]. Based on various experimental studies, which involve the design of pyrheliometer, the circumsolar sky does not contribute considerably to the beam on a clear sunny day. On the contrary, when the sky is hazy or has a uniform cloud cover, the radiation gets distributed, and the contribution of the circumsolar sky to the measurement becomes more significant. Some typical applications of pyrheliometers are material testing research, scientific meteorological observations, climatic observations and assessment of the efficiency of solar collectors and PV devices.

2.4.4 Earth observing satellites: With an increase in the usage of alternative energy resources, the demand for accurate estimates of down-welling surface shortwave radiation (DSSR) for building extensive solar radiation directory and generating short-term solar forecasts, has increased [18]. Despite the expansion in the number of ground station networks globally, the accessible database about solar radiation is not adequate enough, as far as spatial distribution and wholeness are concerned. As a result, satellite-derived DSSR is an essential means for performing analysis and forecasting. Satellites can monitor the state of the atmosphere and cloud-cover on a near-continuous temporal basis and spatial resolutions that cover an area of various kilometers [19, 20]. Satellite data provides information about cloudiness, which in turn provides information about irradiance. Satellite-derived DSSR, in combination with CMV, can bring remarkable improvement in forecasting power for time period of few hours. Additionally, there is an improvement in the cloud fields predicted by the NWP models, because of direct cloud assimilation that is detected through satellites [21, 22].

Meteorological satellites carry sensors which point towards the ground. This enables them to have a bird eye view of the entire globe from space. There are two types of meteorological satellites characterised by their orbits, namely geostationary and polar-orbiting satellites. Geostationary satellites are the ones that revolve around the Earth at altitudes of about 36,000 km and are stationary with respect to the rotating Earth. Consequently, with the help of their infrared sensors and visible-light, they can record and broadcast images of the Earth's hemisphere regularly. Several geostationary meteorological spacecraft orbit the earth with certain objectives. For example, the United States has geostationary operational environmental satellites (GOES), Europe has Meteosat geostationary satellites in operation.

Polar-orbiting meteorological satellites orbit the Earth at an altitude of 850 kms (530 miles) in a south to the north direction or vice versa. In their continuous flight, these satellites pass over the

poles. They can observe any place or location on the earth twice a day, with the same general lighting conditions. The resolution of these satellites are far exceeding than their counterparts as they are very near to the Earth.

2.4.5 Sky-cameras: With the introduction of low-cost digital photography, there has been a paradigm shift in the methods used for sensing the atmosphere of the Earth. The images obtained from geostationary satellites form the primary source of visual information describing the atmosphere of the Earth. However, they often suffer from poor spatial and temporal resolutions. This makes them less significant for locations where the local factors highly govern the weather phenomena, e.g. tropics. In such a scenario, more detailed local information about cloud formations is provided by high-resolution images. These cameras capture the entire sky periodically from the ground, making use of hemispheric lenses. Images obtained from these cameras have better resolution than those obtained from satellites.

Additionally, the upward-facing orientation of sky cameras makes it feasible to capture the low-lying clouds [23]. Cloud motion and distribution is the most critical element creating fluctuation of solar irradiance at ground level [24]. Most of the very short-term irradiance forecasting methods make use of satellite images [25]. However, intra-hour forecast cannot make use of satellite images due to their limitations in their spatial and temporal resolutions. This issue can be solved by using ground-based sky imagers to capture images with high spatial and temporal resolutions for veryshort-term forecasts [26]. However, most of the commercial sky imagers are costly, restricting their use in low-cost applications.

2.4.6 PV array system: Measuring the irradiance using an array of PV systems is another widely used technique for irradiance forecasting [27]. Using sensors that are ground-based eliminates the need for individual estimates of the cloud properties and helps in directly inferring the PV power output from PV systems. This is an advantage of using PV arrays over NWP models or techniques that are satellite based, which need RTMs for converting the images from satellites or the cloud data to a measure of surface-level plane-of-array irradiance. Using PV power measurements for irradiance can eradicate the need for an exclusive network of irradiance sensors resulting in automatic accounting of the incidence angle and temperature effect on solar forecasts. The authors in [4, 28, 29] have used around 80 rooftop/distributed PVs as a sensor network for measuring the solar irradiance. This network covers a 50 km square area in Tuscon, Arizona State, having an average nearest neighbour spacing of 3 km, which provides better spatial and temporal resolution when compared to the current operational forecasts based on images obtained from geostationary satellites.

2.4.7 Shadow cameras: Persistence forecasts cannot predict some of the parameters in the PV plants (like ramp rates) which might lead to uncertain validation results. Based on ground measurements, these results show extrapolation to any existing field [30]. This extrapolation is a result of similarity in spatial and temporal variability for the majority of situations, short time periods (up to 30 min) and areas of a few km². If the comparison is made for its reference station and not for spatially resolved reference irradiance maps, very different behaviour is shown by the persistence forecasts. Owing to these facts, validations of nowcasting systems must be interpreted very carefully. The system of shadow cameras introduced in [30] overcomes such issues as they provide reference irradiance maps rather than measurements from the ground. This enables the spatially resolved benchmarks of irradiance maps obtained from sky imagers. With the shadow camera system demonstrated in [30] and around the number of radiometers used for its validation, the PSA is a validation site suitable for benchmark nowcasts. Such an example of validation has been provided by the authors there. The system presented in the above work can calculate spatially resolved DNI, GHI, and GTI maps. The authors in [30] used a system of six shadow cameras,

which takes photos (every 15 s) from 87 m tower located in Spain. An ortho-normalised image called orthoimage is computed using the photos obtained from these six imagers, which is compared with two reference orthoimages and eventually used for deriving spatially resolved irradiance maps. The preparatory work for this system was done in [31]. Apart from benchmarking nowcasts, this system can be used for generating highly resolved irradiance maps for large scales. Moreover, they can also be used to identify fog and snow as well.

2.5 Conversion of irradiance to power output

Conversion of the forecasted irradiance (G_t) to actual power output ($P_{x,t}$) at any time period (t) mainly depends on the PV power rating (P_{rated}), solar cell temperature (T_t), inverter loss (P^{inv}) and ambient temperature (T_t^a). The conversion can be done using the standard equations [32] described below

$$P_{x,t} = \frac{G_t}{1000[\text{W/m}^2]} P_{\text{rated}} * (1 - \eta_T(T_t - 25[^\circ\text{C}])) - P_{x,t}^{\text{inv}} \quad (4)$$

$$T_t = T_t^a + \frac{G_t}{800[\text{W/m}^2]} * (NOCT - 20[^\circ\text{C}]) \quad (5)$$

where x is the PV index. The power conversion can also be done using many commercial PV modelling software tools in the market such as SAM [33], PVsyst [34], PVSOL [35], Helioscope [36] etc. The users can choose the appropriate software depending on the type of analysis, calculation model, hourly/sub-hourly calculation, modelling capability, user friendliness and cost.

3 Methods for solar irradiance forecasting

In Section 2, the various concepts and terminologies involved in solar irradiance forecasting were discussed. This section reviews the state-of-the-art solar irradiance forecasting methods. Solar irradiance forecasting techniques are needed to generate forecasts at different horizons for reliable, continuous, and efficient integration of solar energy to the grid. There are numerous algorithms and methods that are used for forecasting irradiance and they can be broadly classified based on time series and meteorological data.

3.1 Data-driven methods

Data-driven forecasting has been mainly done using statistical models that creates a relationship among the predictors, input, and output variables. Statistical methods can be broadly categorised as linear and non-linear, a brief discussion of which is given below. The authors in [37] has given a detailed description of the same.

3.1.1 Time-series models: These methods are also termed as time series analysis (TSA) methods and can forecast the irradiance by statistically analysing the directions in past time series of the forecast variables, exogenous variables or any other variables [11], based on theoretical foundations and mathematical representations. The main aim of TSA is to determine a model that not only illustrates the time series pattern and analyses how the past values affect the future values and forecast the future values, but also function as a control standard for a variable to the measure the product quality in certain conditions. TSA of data has been performed in various studies like [38, 39]. Time-series models can be further classified as univariate and multivariate methods. The authors of [40, 41] have given a detailed description of the univariate and multivariate time series methods, a brief discussion of which is given below.

A univariate time series is the quantitative analysis of just one variable in the time domain. It is meant to predict (forecast) a variable based on its past observations. The mathematical representation is given as:

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + a_3 y_{t-3} + \dots \quad (6)$$

where y_t represents the actual value of the time series at t , y_{t-1} represents the actual value of the time series at $t-1$ and a_t represents the auto-regressive parameter for y_{t-1} .

A multivariate time series is a quantitative analysis of multiple variables, in the time domain. The predictor variable comprises of two types of data, namely response data and exogenous data. The predictor data structure is a matrix representing one path of observations. Suppose there are T sample times and n time series. To create a variable representing one path of response data, include the data in the form of a T -by- n matrix:

$$y_t = \begin{bmatrix} y(1,1) & y(2,1) & \cdots & y(1,n) \\ y(2,1) & y(2,2) & \cdots & y(2,n) \\ \vdots & \vdots & \ddots & \vdots \\ y(T,1) & y(T,2) & \cdots & y(T,n) \end{bmatrix}$$

Some of the popularly used univariate and multivariate time series models are discussed below.

Persistence model: This model is the simplest of all the TSA models. It is otherwise called as the “naïve predictor” and is used as a standard benchmark by other methods. It is also a straightforward and common model that is often referenced by solar and wind forecasters, especially for short-term forecasting. This model assumes that the future conditions are the same as past conditions, where the last observed GHI persists at the forecast horizon. The accuracy of this forecasting method decreases with increase in the forecast horizon because the cloud cover changes its state with an increase in the time slot, making it unsuitable for horizons beyond 1 h.

Autoregressive (AR) model: AR model is a representation of the random processes where the present value is given as the function of past p values and a stochastic term.

MA model: Moving average (MA) model is a standard method for univariate time series in which the time series is generated through a weighted average of the present and past observations (of one or more prior values).

AR model with exogenous (ARX) inputs: An AR model with exogenous inputs is termed as an AR model with exogenous inputs (ARX). The exogenous inputs are the additional regressors used in this model.

AR conditional heteroscedastic (ARCH) model: ARCH model is a stochastic method which performs heteroscedasticity corrections by estimating the conditional variance of the present error term. In [42], the authors have used the ARCH effect to test the final error series.

AR moving average (ARMA) model: ARMA model is an aggregate of the AR and the MA models [12]. To make the models useful, they need to have parsimony. In the ARMA model, parameterisation is parsimonious [43], making this model useful in the area of irradiance forecasting. This model is commonly employed to auto-correlated time series data and is considered as flexible, having the ability to represent different time series by changing the order. The restriction for using this method is that the time series has to be stationary [44].

AR integrated moving average (ARIMA) techniques: ARIMA model is the reference estimator for predicting GHI [44]. ARIMA models can be applied for non-stationary time series, as this method is modelled by coupling the AR and MA methods, after particularising an appropriate difference of time-steps for removing any trends [41].

Generalised ARCH (GARCH) model: GARCH model is the variance of an AR model with an ARIMA model. The authors in

[45] talk about using a combination of ARMA and GARCH models in econometrics for providing probabilistic forecasts of solar irradiance. This model can carry out point forecasts by making use of just the past measured values of the solar irradiance. The accuracy of these point forecasts is very similar to the accuracy of the machine-learning-based methods.

Seasonal ARIMA (SARIMA) model: Additional seasonal terms are included in the ARIMA model to form the SARIMA model. In [46], the authors have used this method while using kriging as an optimal spatio-temporal prediction method.

Coupled AR and dynamical system (CARDS) model: CARDS model is an aggregate of AR model and a dynamical system model [47], where the dynamical system model which is the resonating model is used for high-frequency data analysis especially solar radiation. The AR models are used to determine the irradiance statistically. CARDS is a perfect combination of these two models, with improved forecasting performance.

Vector AR (VAR) models: VAR model is a simplification of AR models which reflects linear inter-dependencies among several series, like satellite-derived irradiance series, sensor networks etc. The authors in [48] makes use of this spatio-temporal forecasting method, considering that the physical processes that induces spatio-temporal field is not completely arbitrary, and varied methods can be used for forecast generation. It is represented as

$$y_t = c + \sum_{i=1}^p \Phi_i y_{t-i} + \epsilon_t. \quad (7)$$

Vector moving average (VMA) models: VMA model is an extension of the standard VAR process [49]. When the VAR and MA models are combined, the resulting model is known as VMA model. It is represented as

$$y_t = c + \sum_{j=1}^q \Theta_j \epsilon_{t-j} + \epsilon_t. \quad (8)$$

Vector ARMA (VARMA) models: The VARMA modelling technique allows several dependent time series to be modelled together and accounts for both cross- and within-correlations of the series [50]. It is represented as

$$y_t = c + \sum_{i=1}^p \Phi_i y_{t-i} + \sum_{j=1}^q \Theta_j \epsilon_{t-j} + \epsilon_t. \quad (9)$$

VARMA models with a linear time trend: VARMA with a linear time trend is represented as

$$y_t = c + \delta_t + \sum_{i=1}^p \Phi_i y_{t-i} + \sum_{j=1}^q \Theta_j \epsilon_{t-j} + \epsilon_t. \quad (10)$$

VARMA with exogenous inputs (VARMAX) models: VARMAX method is represented as

$$y_t = c + \beta x_t + \sum_{i=1}^p \Phi_i y_{t-i} + \sum_{j=1}^q \Theta_j \epsilon_{t-j} + \epsilon_t. \quad (11)$$

Structural VARMA (SVARMA) models: SVARMA method is represented as

$$\Phi_0 y_t = c + \sum_{i=1}^p \Phi_i y_{t-i} + \sum_{j=1}^q \Theta_j \epsilon_{t-j} + \Theta_0 \epsilon_t, \quad (12)$$

where y_t is the vector representing the response time series variables at a given time t , Φ_i denotes the AR matrices, Θ_j

represents the MA matrices, ϵ_t is a vector representing serially uncorrelated innovations, c denotes a constant vector of intercepts, x_t denotes $r \times 1$ vector which represents exogenous terms at each time t . r is the count of exogenous series. δ represents a constant vector of linear time trend coefficients. β represents a constant matrix of regression coefficients.

In [51], the authors have used an exponential smoothing state space (ESSS) model for forecasting high-resolution solar irradiance time series. This model is used for forecasting the stationary residual series of data sets from Colorado and Singapore. This model is tested using the same set of data for linear exponential smoothing, ARIMA, random walk models and simple exponential smoothing, and found that ESSS performs better than the others in the forecast horizon of 5 mins to 20 mins. Their original Fourier trend model has been benchmarked with other standard trend models using Kwiatkowski–Phillips–Schmidt–Shin stationarity test and residual analysis. In [52], the authors have proposed models based on three time series decomposition to forecast hourly GHI using exponential smoothing for temporal prediction. The first method uses an additive seasonal-trend decomposition as a preprocessing technique before ETS, which reduces the state space and improves the computational efficiency. In the second method, GHI time series is segregated into a direct component and a diffuse component that is later used as forecasting model inputs and the results are recombined through closure equation for getting the GHI forecasts. The third method uses ETS for the cloud cover forecast and forecasts GHI via polynomial regressions. The Exponential smoothing method comprises of a total of fifteen methods that can be categorised as trend and seasonal [51].

3.1.2 Recursive least square (RLS) models: RLSs method is a variant of least squares method that reduces the weighted least squares minimally. This method is appropriate for short-term forecasting. The authors in [39, 45] have discussed the usage of this model in their work.

Automatic relevance determination (ARD) models: ARD is a technique for assigning regularisation coefficients (weights) to input variables under the Bayesian framework. This method finds application in the training of neural networks as well as linear regression fitting. In [53] the researchers have used artificial neural networks (ANNs) to forecast the hourly GHI from 1 h to 6 h ahead. Authors of [54] have also used ARD as a regression technique.

Generalised linear model (GLM): The GLM is a linear regression model, and it gives flexibility to the response for having a non-normal distribution.

Ordinary least square (OLS) models: OLS is a method used in linear regression model for parameter estimation. In this method, the residual sum of squares is minimised and then used by various researchers worldwide.

Least absolute shrinkage and selection operator (LASSO) models: LASSO is a technique for selecting regression predictor and is based on L_1 regularisation. This method is mainly used for selecting spatio-temporal neighbours in sensor networks measuring solar irradiance. It has been used by [55, 56].

Group regularised estimation under structural hierarchy (GRESH) models: GRESH is a hierarchical variable selection method which has been used recently by [55, 57]. It takes into account the penalties of both LASSO and group LASSO.

Multivariate adaptive regression spline (MARS) models: MARS is a non-parametric technique used for flexible data modelling with large dimensions. In this model, the input space is partitioned into multiple zones where each one of the zones has its own regression equation.

3.1.3 Machine learning: Machine learning approaches are widely popular for applications such as optimisation, pattern recognition, control, security, classification, data mining, and forecasting. They build mathematical models based on training data samples and learning, which involves classification and regression. When one talks about machine learning, it involves various method families like linear (e.g. ridge regression), non-linear (e.g. support vector regression, neural networks), tree-/rule-based (random forests, Cubist), Bayesian/boosted/bagged variants, and etc.

Machine learning models can be used to forecast various horizons ranging from intra-hour to day-ahead depending on the spatial and temporal resolution of the data. Due to the flexibility of these models and the recent advances in the field have spawned a large collection of machine-learning-based solar forecasting studies (see [37] for a detailed review). Most of these models estimate coefficients by regressing the features in a data set. On top of these coefficients, machine learning models have specific parameters (called as hyperparameters) as well, e.g. shrinkage parameter in LASSO, number of hidden layers in neural networks or maximum number of terminal nodes in extreme gradient boosting. The choice (also known as tuning) of these parameters have direct impact on the predictive performance, and the forecast practitioners often take empirical approaches for tuning these parameters. However, empirical approaches require expert knowledge on the machine learning model itself and the software package used. For this reason, more automated and globally applicable approaches—e.g. automatic tuning algorithm in the caret package [58] available in R programming language or Bayesian optimisation approach for tuning [59] available in most of the popular programming languages—are of interest to forecast practitioners. Hyperparameter tuning is also case-dependent, which a set of accurate hyperparameters at a location is not guaranteed to be accurate in other locations. These limitations make machine learning models not commonly applicable for operational forecasting. Instead, they are commonly preferred for post-processing of NWP forecasts as described in Section 3.7 due to their flexible nature and ability of modelling non-linear relationships in the data.

Machine learning models are required to be ‘trained’. A complete set of data set is separated into training, validation and test sets. The training set is used to train the machine learning model. The validation set is used for hyperparameter tuning of these models. After the hyperparameters are tuned and the model is trained, a test set is used to evaluate the predictive performance of these models.

A comprehensive comparison study on predictive performance of machine learning models is conducted by [60], where the authors compared the accuracy of 68 machine learning methods for hour-ahead forecasting in 3 sky conditions in major climates in the continental United States. The authors reported that the predictive performance of machine learning models highly vary across climate zones, weather conditions and evaluation periods, and the predictive performance of machine learning models are case-dependent. Also, the authors reported that although no specific model stands out in terms of performance for all-sky conditions, some model families (tree-/rule-based) are reported as more superior in terms of accuracy.

Evolutionary algorithm (EA) is a population-based optimisation algorithm, which employs algorithms inspired by biological evolution. EAs are generally provide reasonable approximate solutions to problems cannot be solved using simple computations. Optimisation algorithms are generally included under this category. These algorithms find themselves useful in cases where it is computationally intensive to determine a definite solution and a near-optimal solution is adequate. EAs never guarantee an optimal solution to a problem, however they can determine a suitable solution to the problem among an array of possible solutions. Some of the standard EA techniques used for forecasting are differential evolution, genetic algorithm (GA), glowworm swarm optimisation, grouping genetic algorithm and particle swarm optimisation.

These methods recognise patterns in the historical data and forecast the solar irradiance accordingly. One of the most common machine-learning models in the solar irradiance forecasting

literature is ANN. These models can change the magnitudes of their biases and weights that make it possible for them to introduce more changes in a time series. According to the study done in [61], which uses ANN for forecasting GHI, ANN can be used generalised and can be applied to any location, even with incomplete data availability. As per [38, 62], ANN models for training processes are termed as poor forecasting models in case of presence of noise or a complicated model structure. One suggested solution to the above-stated problem is segregating the available data of time series into a training set, a validation set, and a testing set. Moreover, out of these, the training process involves the training and validation sets, which is evaluated using the testing data set [63]. A few ANN techniques include wavelet neural network (WNN), time delay neural networks (TDNN), self organising maps, physical hybrid ANNs, multi-layer perceptron (MLP) etc. Although ANN models are preferred due to their applicability to multi-variable problems, it is recommended in [64] to use fuzzy clustering in the ANN models for defining the sky conditions.

3.1.4 Sensor networks for solar forecasting: This subsection mainly focuses on the sensor based methods used for irradiance forecasting. A detailed discussion and summary of the studies using sensor networks is described here.

General description: The leading cause of the fluctuation in solar power fed into the grid is its spatio-temporal variation caused by the clouds [65]. High-speed clouds cause fluctuations in the PV power, which results in voltage fluctuations, eventually leading to grid instability [66]. Thus it is imperative to analyse the CMVs and develop methods and models to determine the same. In order to predict the cloud movements, the data is gathered from geographically distributed sensor networks, satellite cameras, and stationary cameras [67]. Satellite images predict the clouds movement over a broad region for a time period ranging from 0.5 h to 6 h. These images are mainly used for forecasting cloud movements of huge and slow-moving clouds [28, 29, 68, 69]. Irradiance sensors are cost-effective and easily employable when compared to the famous sky imaging devices [70]. Geographically distributed sensor networks with a dense arrangement of sensors and permanent measurement of output power are good enough to provide input data for short-term CMVs with acceptable accuracy [4, 71]. These networks consist of PV power plants and pyranometers. The accuracy of these sensors is improved by the use of stationary cameras, which can be fixed or moving on a tracking system [72, 73]. Various techniques are applied to sensor networks for determining the solar irradiance. The followings are some example studies.

Statistical and artificial intelligence approaches: LASSO is a penalised regression method. For enhancing the prediction accuracy and interpretation of the statistical model, it performs variable selection as well as regularisation. It involves penalising the absolute value of the regression coefficients. LASSO is used for forecasting solar irradiance over a very short time period in a sensor network [56]. From a hugely correlated lagged time series data that is gathered using a thick network of monitoring stations, LASSO can be used to narrow and choose the spatio-temporal neighbours. The forecasting results of this method is better compared to persistence and other univariate time series models due to the existence of highly correlated data that is obtained from the along-wind station pairs. The performance of the LASSO is limited by the observed spatio-temporal correlations. It outperforms for a dense network of sensors, but the performance decreases for less dense networks. LASSO performed better than OLS, ETS or ARIMA models in [56] where a thick network of 17 irradiance sensors, on the island of Hawaii, was used. However, it performed sub-optimally in [74] where a sparse network of 13 irradiance sensors, on the island of Singapore, was used.

Peak matching algorithm (PMA) is a regression technique used to forecast very short-term solar irradiance using a sensor network. In [75], a wireless sensor network system has been developed to enable high-resolution solar irradiance forecasts. The authors state

that the proposed forecasting model can forecast output of a solar plant (around 1 MW) using the data from the sensor network within a particular time horizon. Here, the PMA dynamically compares and matches the local maxima between the solar plant data and the sensor data, and formulates a prediction. The formulation depends on estimating the present time difference between the recording of a cloud event by a sensor and the solar output this event. Unlike neural network methods, which require an extensive amount of data in training, this method does not require any form of training. However, similar to the LASSO method, PMA is not able to forecast at locations that are unobserved. For improving the accuracy of the forecast, the authors in [76], proposed to combine all the individual forecasts done using PMA like algorithms where the final forecast would be arithmetic mean of all the individual forecasts. However, the shortcoming of this methodology is that it will not be able to produce forecasts with fixed horizons as the PMA entirely depends on the travelling distance and speed of the cloud. Readers can refer to [75] to know specifics of the PMA method, its application to the sensor network and evaluation metrics used to analyse the performance of this method.

As mentioned in [46], LASSO and PMA cannot forecast the irradiance at unobserved locations. In order to address this concern, spatial-temporal kriging is used. Methods like ARIMA analyses, ANN-MLP, k-nearest neighbours algorithm, and Bayesian inference do not yield any information on spatial irradiance despite having adequate accuracy [77]. A considerable amount of monitoring stations are required for planning power generation at the grid level. However, due to the limited number of sensors and their uneven distribution, it becomes essential to estimate the spatial-temporal values of solar irradiance. Kriging is a stochastic weighted interpolation method, where the average and covariances (used for calculating their weights) are preserved, and the error estimation is used for the stochastic simulation. In this method, the unobserved locations are predicted using the covariances between an unobserved location and the monitored locations that are based on their time lags and distance. For spatio-temporal prediction, kriging can be considered as an optimal method depending on statistical dependencies. Though the spatial resolution is considered in most of the studies related to spatial-temporal irradiance, data sparsity becomes a significant limiting factor to this approach. Studies in [78] show that after a certain threshold distance, the spatial correlation between two locations converges. Moreover, in the kriging method, the dependence metrics namely dispersion or correlation, has to be selected attentively. When applying kriging methods to data from places with extensive geographical separations, information losses can occur due to overestimating of the threshold distance.

A kriging methodology for solar irradiance forecasting over a large region with time series inputs from a limited number of monitoring stations is proposed in [77]. They have constructed a statistical forecast system using a network of 10 meteorological stations in Singapore, each station having silicon sensors or pyranometers. In [79], a network of 12 irradiance sensors in Tucson, Arizona is used to forecast the irradiance from 1 min to 2 h. It has been shown that spatial or temporal averaging minimises the forecast RMS errors owing to the factor that such forecasts are smoother and have smaller covariance. In [46], a very short-term irradiance forecasting is carried out by applying the kriging method with empirical and fitted correlation matrices on an area with 1 km radius and 17 stations in Oahu island, Hawaii. Here, the parametric correlation functions are constructed for forecasting at unobserved locations (locations void of irradiance sensors). The authors in [80] used spatio-temporal kriging method for modelling irradiance at a random point, where the temporal and spatial decorrelation distances are analytically estimated. The computational size is minimised through parameter shrinkage, showing a 66% betterment over the persistence model. In [48], the authors considered the scenario of Singapore where data from 13 meteorological stations are used for forecasting solar irradiance by spatio-temporal empirical kriging and vector AR models with parameter shrinkage. These methods quantify the long-term correlation threshold distance within solar irradiance random fields. The threshold distance that is estimated can be considered as

an automatic parameter shrinkage tool for spatio-temporal forecasting models.

A spatio-temporal VAR forecast model is used to predict irradiance for spatially sparse sensor network, which is suitable for data fitted with productive experimental results in the time however, inadequate in the spatial dimension [81]. The authors in [82] proposed a system for short-term forecasting in a tropical, highly urbanised environment like Singapore. The proposed model which relies on weather observations and storm alarm system showed improved performance when compared to persistence methods by 5.8 and 5.0%, for 15 min and 30 min prediction horizons, respectively. This method is one among the most common and simple to use models for multivariate time series analysis. For performing spatio-temporal prediction, the kriging method is more efficient with rich data availability in spatial and time dimensions however, not for minimal measurement sites. The forecasts from VAR models are adaptable as they can be made conditional on the possible future paths of specific variables.

The authors in [83] describes a model-tree-based technique for solar irradiance variability forecasting at eight sites in Sacramento valley. The model used mean absolute error (MAE) to assess the predictive ability of the tree. The forecast lead time for the GHI was calculated, which showed both the spatial and temporal variability. The authors in [84] proposed a mixed WNN model for forecasting short-term solar irradiance. This model includes sky CI as an exogenous input to the forecasting model. Six testing data sets were employed having 15 min and hourly data from Singapore for a period of 12 months. The hourly data was gathered from 25 different sensors and combined to estimate the total available irradiance for each hour. The proposed model was compared with ETS, persistence model, ARIMA, and standard MLP-based models in order to prove its efficiency and effectiveness. In [85], a regime-dependent ANN technique for short-range (15–180 min) solar irradiance forecasting has been proposed. It shows lesser forecast error than global ANN or smart persistence, and this can be employed to forecast irradiance variability. In order to identify a cloud regime, a k-means algorithm is used based on observations from surface weather and irradiance followed by training the ANNs for predicting the CI. Here a network of irradiance sensors in Sacramento, California is used.

Cloud-tracking-based approaches: Determining the height of cloud base is critical for short-term solar forecasting algorithms. Methods like radiosondes, stereographic, and ceilometer are not suitable for intra-hour forecasting as they are highly expensive and have low temporal resolution. Satellite images can surmount this deficiency but can provide only the cloud top height using the thermal IR channel. The authors in [86] introduces a cloud shadow speed (CSS) sensor with angular cloud speed from a sky-imager to deduce the cloud base height (CBH). Here, the authors describe the integration of a CSS with angular cloud speed from a sky imager to determine CBH. This is better approach to deduce CMVs from CSS as it has low noise and improved accuracy than the other methods. This method has proved itself better than the one presented by [87]. The authors in [88] have applied time series analysis for forecasting next hour solar irradiance including effects of cloud cover in the areas of Miami and Orlando. In this approach, examination of the seasonal components namely each type of solar irradiance (DNI and GHI) occurs and later removed from the time series of the solar irradiance. ARIMA models are then used to forecast the residuals. In [87], the authors measure CMVs using a network of ground sensors in a 48W PV power plant in Henderson. LCE and CCM methods have been used to determine the CMVs showing a remarkable improvement in the root mean squared error (RMSE) values. The authors in [89] proposed an embedded nowcasting method using cloud speed persistence and validated using a 48 MW PV power plant in south-western Nevada.

The above mentioned professional forecasting systems are accurate for the prediction of CMVs but at the same time are very expensive. The price of such a system is even more than that of a small-scale solar power plant supplying electricity for a household [90]. In [67], the authors have proposed an economical system for forecasting the cloud movement in the time-interval of less than a

minute i.e., predicting the cloud covering the sun in next minute. The cost of this system is less than the cost of a single 270 Watt PV module. In [91], the authors have proposed an economical sensor network, which enables solar irradiance measurement in the interior of the tree canopy, in Ademuz (Valencia, Spain). Here the authors used two types of sensors namely single c-Si cells and a-Si series-connected mini-modules, and they were tested in two different measurement years.

PV power-based approaches: In [28], it has been discussed how rooftop PV can be used for measurement of solar irradiance and PV power output can be interpreted from PV output of the systems used. One of the main advantage of this approach is determining the PV output without inferring the cloud properties. The sample space used here is a network of 80 residential rooftop systems that are scattered over a 50 km x 50 km area in Tucson, Arizona. The ground sensors are placed 3 kms apart, and measurements are taken every 15 min. The analysis of this system has been done in [29], which includes detection followed by correction for underperformance caused by outages, clouds, partial shade, and module orientation. It also identifies the causes of de-ratings in the PV power, which is essential in making hour-ahead forecasts of PV output over a wide geographical area.

The authors in [92] discusses the modelling of solar irradiance smoothing for huge PV power plants by employing a 45-sensor network and a wavelet variability model (WVM) in Northern Arizona University. This approach demonstrates that prediction of irradiance variability using a combination of 25 different sensors is still lesser than the WVM, hence at least 25 sensors would be required for modelling the irradiance smoothing at shorter timescales for a 30 MW power plant in a similar manner to WVM. To model the smoothing of irradiance, WVM employs cloud speed scaling coefficient whose magnitude signifies the range of smoothing. The authors in [4] proposed a method to predict PV power generation for intra-hour forecast horizon (15–90 min) by using the power measurements from a network of 80 residential rooftop PV systems in Tucson, Arizona. The RMS error is improved by 23% for a horizon of 45 minutes.

Various researchers have used sensor networks across the globe to forecast solar irradiance. A handful of the work done has been summarised in Table 3 which highlights important information like location of sensor networks, number of sensors in the network, forecast horizon, best skill etc. It is conspicuous from the Table 3 that using sensor networks, researchers have focused on predicting the irradiance over a wide range of forecast horizon varying from 10 s to a few hours. Few of them have used exogenous data while others have not. Figs. 1–3 show the relationship between forecast skill (FS) score and the number of sensors used in the network and forecast horizon. The number of sensors varies depending on the network used for forecasting, but there is no particular trend shown by the FS depending on the number of sensors, as shown in Fig. 1. It clearly shows that from the lot that the best available skill is 0.8874, which uses 60 sensors (which is neither the minimum or maximum in the given lot). For the maximum number of sensors in the lot, which is 96, there are two skills. So it can be concluded that there is no specific relation between the number of sensors and the FS, but the skill depends on the methodology used. From Fig. 2, it can be seen that most of the sensor networks are used to forecast irradiance in the very-short or short horizons. Fig. 3 is a magnified version of Fig. 2, which shows the skills for horizons from 0 to 180 min, which shows the plot of FS versus forecast horizon. The best possible skill achieved is 0.81 for forecast horizon of 15 min. This is achieved in [80] where the authors have used spatio-temporal interpolation technique called kriging for forecasting the irradiance data. Based on the visual inspections, there is no direct correlation between number of sensors used in the network and FS score. Also, FS score varies depending on the forecast horizon.

Reconciliation: While integrating solar power into the grid reliably and efficiently, different levels (at transmission nodes, distribution nodes, subsystems, inverters) of aggregation employ different forecasting techniques. Forecasting at all these levels is essential for efficient power management, but an aggregation of these

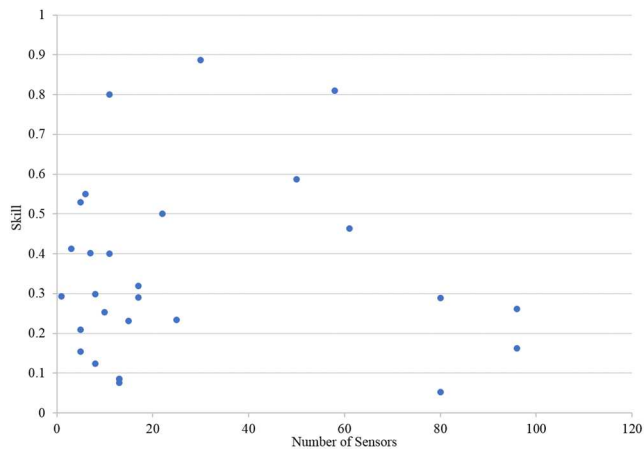


Fig. 1 Number of sensors versus skill

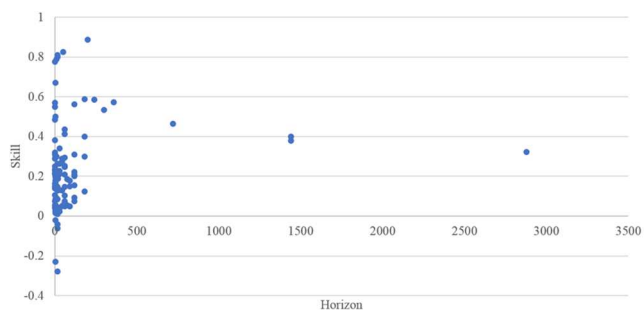


Fig. 2 Horizon versus skill between 0 and 3000 min

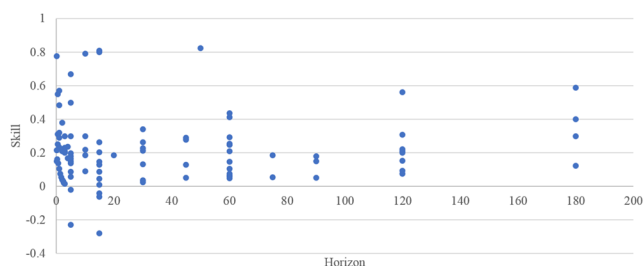


Fig. 3 Horizon versus skill between 0–180 minutes

forecasts is not optimal at different levels due to usage of different forecasting methods in various temporal resolutions. Also, the visibility of different sets of information to different users in the levels of hierarchy prevents them from generating their forecasts. Besides, the predictions done using data collected at higher levels are not necessarily the exact summation of the forecasts done using lower-level data. This is known as ‘aggregate consistency’ problem. In order to produce aggregate forecasts, the most commonly used approaches include bottom-up, top-down, and middle-out approaches [101]. In the bottom-up approach, only the bottom level forecasts are required, and the upper-level forecasts are calculated through aggregating the bottom-level forecasts. In the top-down approach, only the top-level forecasts are required, and the bottom-level forecast is obtained through distributing the forecasts proportionally based on size of the system. In the middle-out approach, the middle-level forecasts are required, and the top and bottom-level forecasts are achieved through aggregation or disaggregation of the middle-level forecasts. However, a consistent aggregation in hierarchy levels is not guaranteed even with the usage of these methods, making them unsuitable for hierarchical solar irradiance forecasting. Moreover, it is tedious to transmit the forecasts that are generated at one level to other levels.

In [102], the authors have explored the state-of-the-art forecast reconciliation methods and applied them for forecasting solar energy in the spatial hierarchy. In this work, the base forecasts obtained across the hierarchy is optimally reconciled by minimising the trace of forecast error covariance matrix. The reconciled forecasts aggregate consistently in spatial hierarchy leading to collaborative decision making, and show significant improvement in the base forecasts at different levels. Using the reconciliation framework avoids the need for any further information besides the base forecasts. Also, reconciliation is not depended on the base forecasting methods. The authors have used this technique for the locations of California (NWP-based and time series-based day-ahead forecast reconciliation over 318 simulated PV plants) and Oahu island, Hawaii (1 min-ahead forecast reconciliation over 17 irradiance sensors). It has been established by the authors that reconciliation brings more accurate forecasts, thus benefiting both the grid operators and PV system owners. This encourages sharing of information in a power grid. An aggregation of hourly/high-frequency time series into various daily/low-frequency time series can result in the construction of a temporal hierarchy. Moreover, many forecasting methods such as theta model, multi-aggregation prediction model, wavelet decomposition model, and empirical model decomposition model are developed

Table 3 Summary of papers on solar irradiance forecasting using sensor network

Reference	Location	Number of sensors	Horizon lower	Horizon upper	Radius	Width	Length	Best Skill	Best skill horizon
[93]	Colorado	5	5	120	130	162	242	0.154	120
[94]	Japan	61	1440	2880	216.645	276.5	333.6	0.463	720
[70]	New York	25	1	5	0.922	1.4	1.2	0.234	1
[46]	Hawaii	17	0.167	5	0.5	1.2	1.2	0.29	1
[56]	Hawaii	17	0.167	5	0.5	1.2	1.2	0.319	1
[89]	Copper Mountain	96	0.333	3	0.9	1.8	0.54	0.162	0.333
[95]	Copper Mountain	96	5	15	0.9	1.8	0.54	0.262	15
[79]	Tucson	15	1	120	25	22.7	46.7	0.231	3
[4]	Tucson	80	15	90	25	51	47	0.053	75
[85]	Sacramento	8	15	180	31.8	45	45	0.124	180
[80]	Sacramento	58	1	15	27	35	44	0.81	15
[83]	Sacramento	8	15	180	31.8	45	45	0.299	180
[96]	Italy	5	180	180	68.5	113	77	0.53	180
[48]	Singapore	13	5	5	19	22	42	0.086	5
[82]	Singapore	10	60	60	19	22	42	0.253	60
[69]	SIRTA, France	1	15	60	0	0	0	0.293	60
[97]	Castile and León	53	15	15	205.69	338.7	233.5	0.587	180
	Castile and León	30	60	360	100	200	200	0.8874	200
[98]	Athens	11	1440	2880	338.75	269.7	621.6	0.4	1440
[99]	California	27	60	180	513.408	512.8	889.6	0.401	180
[100]	New York	12	15	120	2605.124	4217	3060	0.209	120

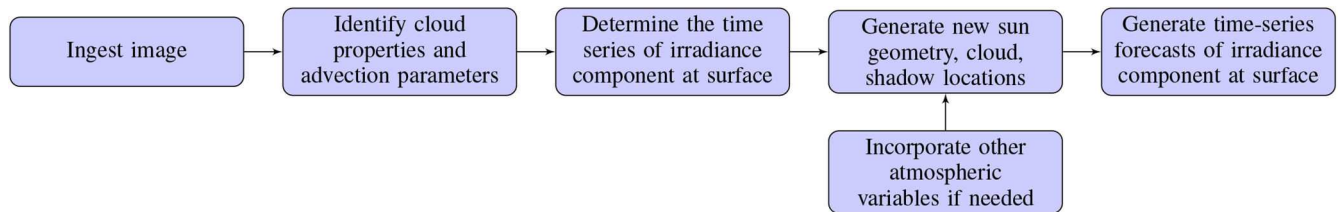


Fig. 4 A sample flowchart illustrating image-based forecast generation

based on multiple decompositions of time series. However, in a temporal hierarchy, the measurements can be consistently aggregated, whereas the forecasts do not aggregate consistently as different models are used for generating forecasts. This can become an apparent issue during forecast selection. Moreover, in these methods forecasts are produced only at the timescale of the original series. Also, modelling errors can occur as the forecasts generated by these time series are not summed up consistently when the initial time series is summed up into a new series that can have lengthened timescales. Through reconciling various forecasts that are obtained at various timescales, both these drawbacks can be mitigated, therefore producing consistent aggregate forecasts. Forecasts can be generated regardless of temporal resolution of input data. More specifically, hour-ahead forecasts can be generated using data with 1-h resolution (1-step-ahead forecasting) or 30 min resolution (2-step-ahead forecasting) or 15 min resolution (4-step-ahead forecasting). Similar to spatial hierarchy, these forecasts generated using different temporal resolutions can constitute a temporal hierarchy, where for example, hourly forecasts aggregate to 2-hourly forecasts, 2-hourly forecasts aggregate to 4-hourly forecasts and so on. Hence, these forecasts should be reconciled in temporal domain for aggregate consistency. In [103], the authors have applied the reconciling technique for forecasting solar irradiance in the temporal domain using hourly data from 318 different sites in California for one year with around 50 million forecasts produced. The temporal reconciliation method has been found to outperform NWP forecasts generated through 3TIER, with a FS up to 0.3.

The authors in [104] proposed ‘sequential reconciliation’ by combining power system spatial and temporal hierarchies so that aggregate-consistent forecasts are generated both in spatial and temporal domains. Independently generated base forecasts are reconciled using two techniques: (i) forecasts are firstly reconciled in space then in time and (ii) forecasts are firstly reconciled in time then in space. Both techniques improved the accuracy of base forecasts.

The main merit of reconciliation methods is that they are data, hierarchy, and method independent. Hence, they are globally applicable. Also [105] reported that reconciliation accuracy depends on the accuracy of base forecasts, where more accurate base forecasts result in more accurate reconciliations.

3.2 Image-based forecasting models

Clouds are variable in both time and space, thus deducing their existence at a specific time becomes a crucial task for forecasting solar irradiance data and devising appropriate methods. Horizontal advection causes cloud motion, which strongly influences the temporal change of cloud structures [106]. The CMV method uses the data from satellites and ground-based sky imagers to track the cloud movements remotely [11]. The images of future cloud cover can be predicted using CMVs, which describes the advection of present clouds, which in turn is used for forecasting the solar irradiance.

3.2.1 Satellite images: The first step for using satellite images in deriving CMVs is the identification of stationary features of clouds within a tolerance limit. Then a tracer within the target image is selected, which corresponds to the easily traceable cloud features. These tracers are expected to be consistent, and spatial coherence analysis tests check the feasibility for consistency. Eventually, a pattern matching algorithm is used to track the tracers, which counterparts an element from the goal region of one image to a

quest region in the subsequent image [11]. The standard method used is the maximum correlation method that deduces the maximum cross-correlation coefficient from sequential images for calculating the shift vector of a selected element.

Other pattern matching processes that are employed for cloud movement tracking are: (i) sum of absolute differences, (ii) sum of square differences and (iii) cross-correlation [107]. The automatic cloud tracking algorithm is another technique where the images from the satellites are applied to the standard pattern correlation coefficient. In [108], a method based on Meteosat satellite images for solar irradiance forecasting was proposed as a basis for solar power forecast. Here, the authors examined and compared different methods for deriving motion vector fields that are obtained from data in Meteosat, and subsequently used those methods for forecasting solar irradiance upto a few hours ahead. In [6], the outcome of forecasted irradiance is shown, which are obtained using the images of the GOES with a similar process emulated after [109].

In the absence of high-quality ground-based measurement, satellite images and satellite-derived irradiance data are often used in forecasting models especially in numerical weather models. However, it should be noted that satellite-derived irradiance data have uncertainties and often show systematic bias mainly due to errors in aerosol optical depth and cloud detection. With this in mind, satellite-derived data should be used with cautious. A recent study compared machine-learning-based irradiance forecasts independently generated using univariate satellite-derived data and ground-based data [110]. Two sets of forecasts showed substantial differences in terms of sharpness and calibration.

3.2.2 Ground-based sky images: Ground-based sky images possess higher spatial and temporal resolution when compared to the satellite data. This leads to the likelihood of apprehending abrupt irradiance changes, usually mentioned as ramps. NWP models and satellite images are not suitable for intra-hour predictions as high spatio-temporal resolution is needed. For intra-hour forecasts, it is better to use ground observations. Total sky imager bridges this predicting gap [106]. The maximum probable forecast horizon rely upon the velocity of the cloud and is governed through the time slot for the cloud that is monitored to cross the region under investigation. This time window is determined by the spatial extension of the monitored cloud scenes in combination with cloud velocities. Imaging instruments that are ground-based using cloud cover indices [111], CMVs [112] and automated cloud classifications [113] lack in spatial coverage, and their performances are site-specific. On the contrary, images from the satellite have higher spatio-temporal resolution ensuring easily observable cloud structures.

Fig. 4 illustrates a sample flowchart of generating irradiance forecasts from an image.

3.3 NWP models

NWP models provide details of atmospheric conditions in time-domain using physical equations and generate forecasts for various variables including but not limited to solar irradiance, wind speed and temperature. NWP models for forecasting are based on the atmospheric physics, thus differing from the statistical methods that are based on empirical relationships. NWP is very much applicable for forecasts in the future that are more than a few hours or days such as intra-day forecasts and day-ahead forecasts. The steps involved in this method are, forecasting through assimilating initial observational data, propagating these conditions forward,

and error correction through statistical post-processing based on past performance [114]. The irradiance forecast errors in NWP is mainly due to the reason that NWP models do not completely resolve the small-scale boundary layer clouds which leads to parameterisation of their representation. Ensemble forecasting is very common to quantify the uncertainty in NWP models, which are perturbations of initial conditions of an NWP model. Ensembles are crucial to understand case-dependent information on forecast uncertainty [115]. Thus, most of the meteorological centres issue ensemble forecasts as a part of their operation suites today since the first operational ensemble forecasts were issued in 1992 [116].

NWP models are computationally expensive to run, and most of these models run only twice in a day (more specifically, 00:00 and 12:00 UTC) [116]. These limitations make NWP models not operationally applicable for intra-hour and intra-day forecast horizons. For such short-term forecast horizons, data-driven models are mostly preferred [117]. NWP models systematically overestimate (bias) the irradiance due to the modelling uncertainties. For this, irradiance forecasts that are based on NWP are generally complemented with a post-processing activity, like model output statistics (MOS), Kalman filter, temporal interpolation, spatial averaging etc. Using statistical post-processing also scales down the output of NWP which is usually unrefined in spatial and temporal resolution for conventional PV related applications, to refined temporal levels. These changes included in the post-process can later be corrected using statistical procedures [118].

Few of the NWP models include weather research and forecasting, regional deterministic prediction system, regional atmospheric modelling system, north American mesoscale, national digital forecast database (NDFD), mesoscale atmospheric simulation system, integrated forecasting system, global forecast system, advanced regional prediction system, and ensemble MOS. Despite their benefits, NWP forecasts are computationally demanding and usually must be run on supercomputers. Furthermore, atmospheric processes are complex, and as NWP models approximate them crudely, forecasts occasionally go awry in ways that are difficult to diagnose, understand, and prevent [114].

3.4 Hybrid models

The hybrid models are formed by combining various methods for improving the prediction accuracy of solar irradiance. These methods are mainly formed to overcome the shortcomings of each method and improve the forecast accuracy. Some of such hybrid methods proposed by researchers [119–122] have outperformed the individual forecasting (see [123] for a detailed review on hybrid models). A hybrid model was developed by [119, 121], which was used to forecast total daily solar irradiance sequences. They have used a combination of ANN and wavelet analysis. In [120], a specially designed ANN with training algorithm, in combination with wavelets, which is based on diagonal recurrent WNN has been used for forecasting the next day hourly values of GHI. To these models, various groups of meteorological observations are fed as inputs. A hybrid model combining both TDNN and ARMA has been used in [122] to enhance the accuracy of the forecast by assuming the existence of linear and non-linear components in the daily irradiance series. The linear component was fitted using the ARMA model, and for deducing the non-linear pattern, TDNN model was used. By combining these two models, the intricate features and strengths of each model are extracted, leading to improved accuracy than the TDNN or ARMA models individually. Fig. 5 shows a flowchart for the hybrid models.

Overall, forecasting model selection is highly dependent on the forecast horizon of interest as different models stand out in different horizons in terms of predictive performance. Persistence models are accepted as superior in very short-term forecast horizons (e.g. up to a few minutes). Image-based models are mostly superior in intra-hour and intra-day forecast horizons, where sky/shadow images are mostly preferred for intra-hour forecasting, whereas satellite images are often used for a forecast

horizon ranging from 30 min to 6 h. The applicability of data-driven models varies depending on the spatial and temporal resolution of the data used. However, they are mostly advantageous for forecast horizons ranging from 30 min to 3 h. Finally, NWP models are well accepted for 6–48 h ahead forecasting. The comparison of forecasting models is not straightforward due to the inconsistent performance of models. Thus, recently, a consensus started emerging that predictive performance of forecasting models (especially data-driven models) are case-dependent and not consistent at all locations due to the differences in local climatology and topography [60, 124].

4 Data sets used for sensor networks

In this section, some data sets commonly used by researchers for the sensor networks in different countries like US, Germany, Singapore, and Spain are discussed and some of the open source data links are provided.

4.1 NDFD, New York

NDFD is an experimental product from the US National Weather Service (NWS) which provides a gridded forecast for the entire country. NWS is a US agency responsible for forecasting weather, issuing hazardous weather warnings and any other weather-related issues to the public and organisations. It gathers and analyses more than 76 billion observations releasing around 1.5 million forecasts and 50,000 warnings annually, with the help of 13 river forecast centres, nine national centres, 122 weather forecast offices, and other support offices. The forecasting work of local and regional NWS offices along with the national centres for environmental prediction (NCEP) assemble the NDFD. At first, the local forecasts which is a byproduct of national model outputs, mesoscale runs and inputs from the humans are produced, and later merged and assembled on a national grid. NDFD is a suite of gridded forecasts of sensible weather elements namely, cloud cover, maximum temperature etc. In order to create a smooth mosaic of digital forecasts, the NWS offices are combined with NCEP. The meteorological development laboratory has contributed to the advancement of the NDFD and its companion project, the national digital guidance database in [125].

4.2 Oahu, Hawaii

This network provides the irradiance data from the national renewable energy laboratory, Oahu network. The network comprises of 17 radiometers. This network stretches over latitudes 21.30812°N to 21.3533°N and longitudes $158.07785^{\circ}\text{W}$ to $158.08700^{\circ}\text{W}$. From each station, the GHI data with 1 s resolution is available [46]. This network makes it feasible to observe highly correlated time series [56]. The authors in [126] gathered data from this network for broken clouds over 13 days and presented the likelihood of observing highly correlated time series of this network found in [127].

4.3 Jülich, Germany

As a part of HOPE campaign, from 2 April to 24 July 2013, a high-density network consisting of 99 autonomous pyranometer stations was operated around Jülich, Germany. This network was spread across a spatial domain having coverage of 50.85°N – 50.95°N and 6.36°E – 6.50°E , which is approximately $10\text{ km} \times 12\text{ km}$ area. These stations recorded the global radiation continuously using a silicon photodiode pyranometer (model: EKO ML-020VM) with 10 Hz resolution [128]. The Global Positioning System module embedded on the data acquisition board of each station provides an accurate time reference. Standardised atmospheric measurement data (SAMD) has established a structure of distributed data servers with a shared web portal hosted by the integrated climate data center (ICDC), Hamburg as the global entry point. The data are archived at the ICDC as ‘SAMD’ and the details are found in [129].

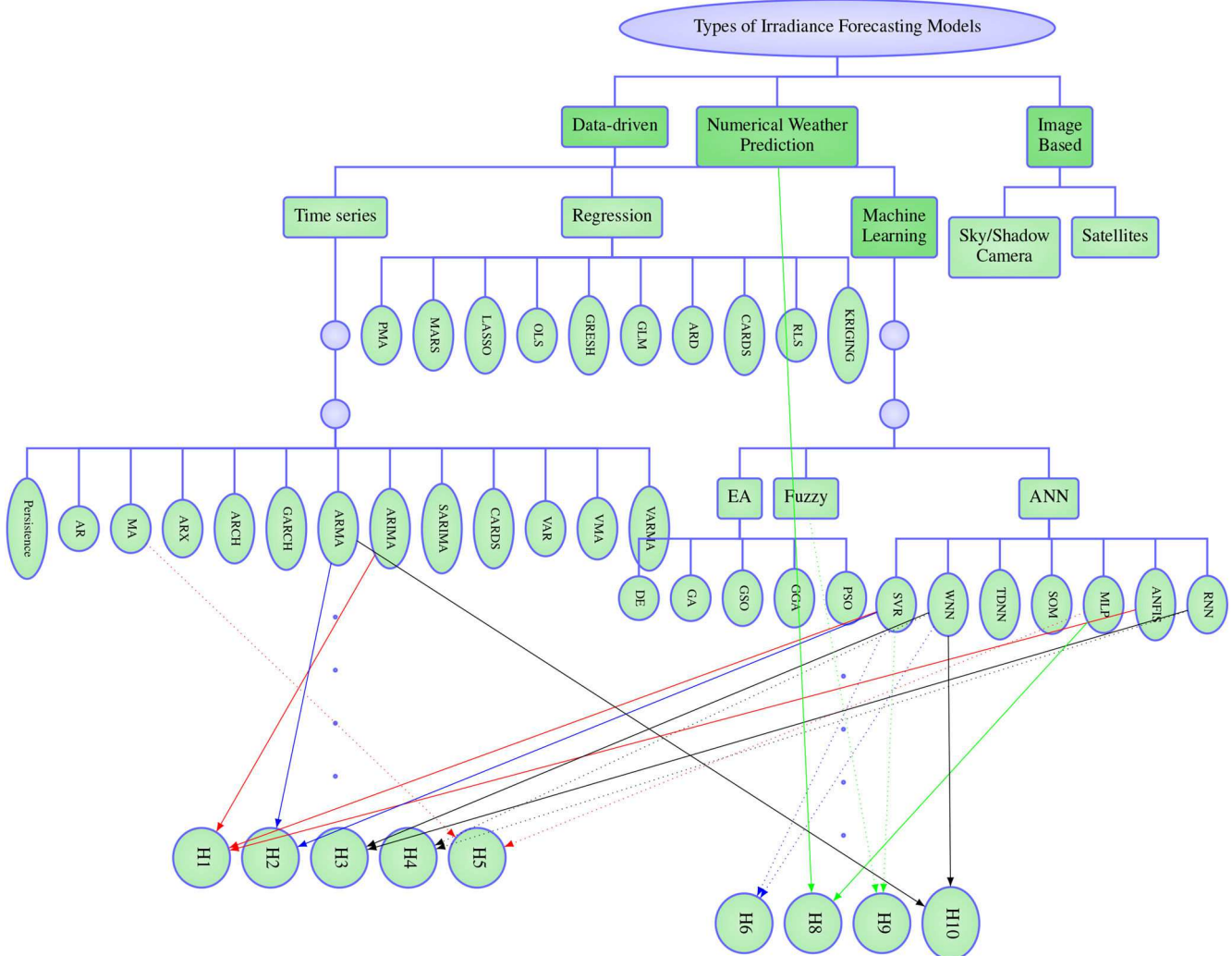


Fig. 5 Irradiance forecasting and hybrid methods

4.4 Tucson, USA

The calibrated sensor network of NREL solar resource and meteorological assessment project (SOLRMAP) site at the University of Arizona is another site attracting the interest of researchers. This site is located at the latitude of 32.23°N and longitude of 110.96°W and elevation of 786 meters AMSL. The sampling rate for the data to be collected is 1 s and is transmitted to the database every minute. Though researchers use both commercial pyranometers and photodiodes, a commercial sensor namely, Apogee SP-212 is used for evaluating forecasted irradiance and complies with the calibrated sensor to within 2% on average on clear days. The data for this network is publicly available in [130]. In [79] intra-hour forecasting method that uses data from a real-time irradiance monitoring network is proposed. The forecasting method used here depends on the network of sensors that sample the GHI at several site. The network consists of 12 irradiance sensors, plus three rooftop PV power systems and one calibrated commercial sensor. The calibrated sensor is part of an NREL SOLRMAP site at the University of Arizona.

4.5 Colorado, New York (NY), USA

The authors in [93] use ground measurement data from NY. For their network design, they have used the gridded satellite derived data from the State University of NY. This data set contains hourly estimate irradiance information for a stretch of 10 km (including global, direct, and diffuse irradiance) for all the states in the US. The state of Alaska is excluded here as the satellite cannot determine the cloud cover data for the year 1998 to 2005. This satellite-derived data set was constructed with GOES imagery using [131] model. Many researchers have used this data set, and

its accuracy is adequate for any research in the field of sensor networks. The data for this network is available in [132].

4.6 Singapore

In [133], the author has used a network of 24 stations in Singapore (41.8 km ENE–WSW and 22.5 km SSE–NNW). The nearest station pair has a separation distance of 0.3 km, and the farthest pair is 39.1 km apart. Frequently calibrated crystalline silicon sensors, that are certified by the Fraunhofer ISE CalLab, with a $\pm 5\%$ uncertainty, are deployed. The network samples GHI are valued each second and logged as 1 min averages. In [82], the authors have considered a network of eight measurement stations. This network in Singapore contains several meteorological stations and was deployed by the Solar Energy Research Institute of Singapore (SERIS) of Singapore [134] and the data is available on request.

4.7 Castilla and León, Spain

The region of Castilla and León comprises nine different provinces and is located in the northwest of Spain. Castilla and León has an area of 94,193 km², about one-fifth of the whole of Spain. Researchers use various networks for their purpose of study. The authors in [97] have used data from a network of 53 ground stations, whereas in [135] the authors have used a network of 12 sensors for measuring the solar irradiance.

Ground-based irradiance data collection is often performed locally. The measurements from each sensor are collected using data loggers, and subsequently, the collected data are transferred and stored in servers. One example of such data logging system is explained in [136], where the data are collected using Campbell

Scientific CR1000 datalogger and are automatically transferred to a MySQL database. Collected raw irradiance data should be preprocessed before any subsequent forecasting application, especially for data-driven forecasting approaches. Common preprocessing practices involve data quality checks, imputing missing or rejected data points due to sensor/communication failures and/or quality checks, and filtering daytime data. With the recent advances in sensing technologies, more environmental data become available for scientific or industrial use. However, the quality of data should be assured as the spurious data may propagate into subsequent applications and spoil their quality as well. The quality of data can be maintained in two stages: (i) before data sensing, where well-motivated sensor maintenance and calibration practices are conducted by a site engineer and (ii) after data sensing, where the data is quality controlled using well-defined data quality control checks. For ground-based irradiance data, a six-step quality control procedure is defined (see [137, 138] for more details) for GHI, DNI and DHI components of solar irradiance. The procedure includes the following quality tests: (i) physically-possible limits, (ii) extremely-rare limits, (iii) comparisons, (iv) climatological limits, (v) climatological comparisons and (vi) cloud enhancement detection for temporally aggregated data. Among these six tests the physically-possible limits or extremely-rare limits tests are the most common ones. After the quality checks, the missing and rejected data points should be imputed as most of the forecasting models cannot handle missing data points. Numerous techniques have been proposed to impute irradiance data gaps, including but not limited to simple approaches, such as persistence or replacement with satellite-derived irradiance or complex approaches, such as spectrum analysis. The accuracy of imputed data is crucial as site-specific irradiance characteristics should be preserved to generate accurate forecasts. In [139], the authors benchmarked the performance of various imputation techniques for irradiance data. Linear and Stineman interpolations, and Kalman filters are suggested to impute missing or rejected irradiance data for various temporal resolutions. After the temporarily complete irradiance data are generated using an accurate imputation technique, it is common to filter daytime irradiance data, which is mostly defined as solar zenith angle $Z < 85^\circ$, before any subsequent applications.

5 Error metrics

The accuracy of solar irradiance forecasting can be evaluated using a set of error metrics. The deterministic solar forecasting metrics can be broadly categorised as ramp characterisation metrics, uncertainty quantification, statistical metrics for different time and geographic scales, propagation metrics, and economic metrics [140]. Statistical metrics includes RMSE, Pearson's correlation coefficient, maximum absolute error, normalised RMSE (NRMSE), MAE, MBE, Kolmogorov–Smirnov test integral (KSI) and OVER metrics. FS is another important metric used for defining the accuracy and degree of association with the predicted observation. It measures how a given forecast is superior over simple historical observations. Few of the popular and widely used error metrics are detailed as follows, where p_i denotes the actual value at the time step i , $\hat{p}_i$ represents the subsequent value approximated by a forecasting model and N denotes the number of data points for evaluation.

5.1 Mean absolute error

This error metric is a global error metric and is suitable for the evaluation of uniform forecast errors [140]. MAE is widely used for regression problems as well as for the evaluation of forecast performance in the renewable energy industry, measuring the overall forecast accuracy. Smaller the value of MAE better is the forecast. It is not advisable to use MAE in systems where extreme events are a matter of concern because a large number of very small errors is more alarming than a small number of large errors [141]. The MAE is represented as

$$\text{MAE} = \frac{\sum_{n=1}^N |\hat{p}_i - p_i|}{N}, \quad (13)$$

5.2 Mean bias error

MBE is appropriate for evaluating the average forecast bias. The larger the value of MBE, the larger is the forecast bias. A positive value of MBE signifies over-forecasting, while a negative value signifies under-forecasting. This enables the grid operators to assign resources in a more efficient way of compensating the forecast errors. The MBE is mathematically represented as

$$\text{MBE} = \frac{\sum_{n=1}^N (\hat{p}_i - p_i)}{N}. \quad (14)$$

The only limitation of using MBE is that it does not provide a good representation of the entire range of forecast errors, for example, the same MBE value could depict various error distributions, where some may be more favourable than others [76].

5.3 Root mean square error

RMSE is appropriate for assessing the complete forecast accuracy while penalising out-sized forecast errors in the order of their square [140]. To get the relative error instead of an absolute error, RMSE value can be normalised either using the capacity value of the solar plants (which have been analysed) or the mean of actual data points, and percentage error can be obtained by multiplying NRMSE with 100 [142]. The forecast error is directly proportional to the RMSE or NRMSE values. This metric likely aggravates large errors in forecasting due to the squaring of each error term [141]. This effectively weighs large errors more compared to the small errors. It is beneficial for evaluation of the complete performance of the predictions, particularly when an adverse situation is a matter of concern. Mathematically RMSE is represented as

$$\text{RMSE} = \sqrt{\frac{\sum_{n=1}^N (\hat{p}_i - p_i)^2}{N}}. \quad (15)$$

5.4 Kolmogorov–Smirnov test integral

KSI estimates the numerical resemblance among the forecasted and actual solar power. This metric is suitable for comparing estimates for longer time periods and for determining the similarities in distributions of forecasts and actual values of forecasts for the time period under consideration [143].

$$\int_{-\infty}^{\infty} |F_n^y(x) - F_n^{\hat{y}}(x)| dx, \quad (16)$$

where $F_n^y(x)$ and $F_n^{\hat{y}}(x)$ represent empirical cumulative distribution functions of measurements and forecasts, respectively.

5.5 Forecast skill

A FS is a parameter to quantify the performance of forecasts over a reference model. It is defined as the ratio of forecast model RMSE to the reference model RMSE [26], where the reference model is mostly persistence and clear-sky adjusted persistence model. The positive values of the FS, therefore, signify that the forecast method under consideration is better when compared to reference forecast, whose maximum possible value can go up to 1. The FS is given as

$$\text{FS} = 1 - \frac{\text{RMSE}_f}{\text{RMSE}_r}, \quad (17)$$

where RMSE_f and RMSE_r indicate the RMSE errors for forecast and reference models, respectively. When FS is 1, the solar forecast

is considered to be perfect, but the uncertainty is as large as variability when FS is 0. The forecast models that improve on reference are characterised by values of FS between 0 and 1. Higher the value of FS, better is the FS [26]. The popularity of FS is clear from the fact that it has been used by quite many famous researchers in this field, as mentioned in [48, 79, 144, 145].

5.6 Ramp and time alignment metrics

In [146], the authors discussed the limitations of conventional error metrics, and proposed two new metrics, namely, temporal distortion mix (TDM) and ramp score, to address the abilities to forecast lag and ramp. TDM is built on the idea of temporal distortion index [147], which is based on a dynamic time warping. Ramp score is based on the idea of swinging-door algorithm, and it measures how well the forecasting model predicts the ramp event in the solar data. The reader is referred to [146] for more details about these two recently proposed metrics.

5.7 Metrics to evaluate sky-imagery-based forecasts

Evaluating the accuracy of sky-imagery-based forecasts is often conducted by using metrics described above after converting images to time-series forecasts. However, the authors in [148] used 3 different metrics to evaluate prediction of the state of the visibility of the sun—namely, Precision (PR), Recall (RE), and F_β score. PR refers to how well predicted occlusions match actual occlusion, whereas RE refers to how well the actual occlusions are predicted. F_β score evaluates these two criteria in a single metric. The metrics are defined as follows:

$$PR = \frac{\text{hits}}{\text{hits} + \text{false alarms}}, \quad (18)$$

$$RE = \frac{\text{hits}}{\text{hits} + \text{misses}}, \quad (19)$$

$$F_\beta = (1 + \beta^2) \times \frac{PR \times RE}{\beta^2 \times PR + RE}, \quad (20)$$

where ‘hits’ are defined as where the prediction and the state are occluded, ‘false alarms’ are defined as where the prediction is occluded, but the state is free sky, and ‘misses’ are defined as where the prediction is free sky, but the state is occluded. Authors in [148] used F_2 score by assigning $\beta = 2$ to emphasise the RE values more.

6 Conclusion

This is a review study based on solar irradiance resource and solar forecasting with a focus on use of sensor networks for forecasting. The authors have presented an overview on solar irradiance resource, radiometers, sensor network data sets, forecast error metrics, and a detailed review of the methods used for forecasting the irradiance in different time horizons. The review shows that solar forecasting techniques can be categorised into three main category: (i) NWP, (ii) image-based methods and (iii) data-driven methods. NWP are mostly appropriate for 6–48 h ahead forecasting. Sky/shadow cameras and satellite-based Earth imagery are the main image-based methods, where the former is preferred for intrahour forecasting; whereas the latter is more appropriate for 0.5–6-h-ahead forecasting. Finally, data-driven methods are applicable to a wide range of forecast horizons based on the input data that ‘train’ the method. They are appropriate from intrahour to even month-ahead forecasting.

High radiometer deployment costs and the difficulties maintaining these radiometers, regarding motivated data quality and calibration practises, restrict the availability of high quality ground-based irradiance measurement networks. Hence, modelled irradiance (e.g. reanalysis) and remote-sensed irradiance (e.g. satellite-derived) products are in increasing demand. Reanalysis data are available with many atmospheric variables along with irradiance over decades. The data are generated using atmospheric

physics and data assimilation techniques, and are lagged (available up to a couple of months) as reanalysis models the past atmospheric conditions. Owing to these features, reanalysis data are more appropriate for climate studies. Satellite-derived irradiance data is generated using Earth images so that ground-based irradiance values are approximated, and the data are available up to near real-time. Considering the recent advances in remote-sensing technology and real-time availability, the researchers and forecasters tend to use remote-sensed products for forecasting tasks. However, this product has model-led systematic errors (biases) compared to ground-based measurements, and it is highly expected that the bias propagates into subsequent applications. For that reason, recent studies focus on improving the quality of the products using statistical and machine learning models. The most common technique is regressing data of these products to high-quality ground-based measurements using multiple features, e.g. CSI, solar zenith angle, temperature, relative humidity and so on, so that the bias is mitigated and the overall quality of these products are improved. Improved modelled or remote-sensed data are used for subsequent applications.

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